A BASIS OF WEAKLY CONTRACTIBLE OBJECTS CONTROLS EXACTNESS, PRODUCTS AND DERIVED LIMITS OF MODULES ON A SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full for all seven parts: the criterion on the basis with local lifting, the limit-dimension input, the surjectivity computation, the product exactness via the abelian-group lemma, and the derivation of the two derived-limit descriptions from the defining triangle.

Editorial note. The label proposition-enough-weakly-contractibles names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own proposition environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source sites-cohomology.tex –label proposition-enough-weakly-contractibles –start 13800 –end 13918 –local-id 3547 returned RESULT=OK (L3547 free, occupancy FREE).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Weakly contractible objects. Same chapter, the lemma recording that sections over a weakly contractible object form an exact functor with vanishing higher cohomology. Supplies the exactness used in parts (1) and (4) and the surjectivity in part (3).
- Limit dimension of the site. Same chapter, the lemma relating the vanishing of derived limits to the limit dimension d. Supplies part (2) with $d = 0$.
- Exactness of finite products of abelian groups and products in the derived category. Homology, the lemma that a finite product of exact sequences of abelian groups is exact, and Derived Categories, the lemma describing products in the derived category, together with the definition of the derived limit. Supplies parts (4), (5) and the distinguished triangle used for parts (6) and (7).

Source reference. Original definition 08TC: sources/stacks-chow/tags.txt, source lines 11160–11160. Prerequisite pointer invoked at proof line(s) 13899; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 07KC: sources/stacks-chow/tags.txt, source lines 9652–9652. Prerequisite pointer invoked at proof line(s) 13894; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 060K: sources/stacks-chow/tags.txt, source lines 7729–7729. Prerequisite pointer invoked at proof line(s) 13889; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D64: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5440–5440. Prerequisite pointer invoked at proof line(s) 13854; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 090K: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13773–13773. Prerequisite pointer invoked at proof line(s) 13839; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0940: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5016–5016. Prerequisite pointer invoked at proof line(s) 13897; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 00WM: sources/stacks-chow/tags.txt, source lines 1188–1188. Prerequisite pointer invoked at proof line(s) 13863; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that every $U \in \mathcal{B}$ is weakly contractible and every object of $\mathcal{C}$ has a covering by elements of $\mathcal{B}$. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Then*

1. *A complex $\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3$ of $\mathcal{O}$-modules is exact, if and only if $\mathcal{F}_1(U) \to \mathcal{F}_2(U) \to \mathcal{F}_3(U)$ is exact for all $U \in \mathcal{B}$.*
2. *Every object K of $D(\mathcal{O})$ is a derived limit of its canonical truncations: $K = R\lim \tau_{\geq -n} K$.*
3. *Given an inverse system $\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$ with surjective transition maps, the projection $\lim \mathcal{F}_n \to \mathcal{F}_1$ is surjective.*
4. *Products are exact on $Mod(\mathcal{O})$.*
5. *Products on $D(\mathcal{O})$ can be computed by taking products of any representative complexes.*
6. *If $(\mathcal{F}_n)$ is an inverse system of $\mathcal{O}$-modules, then $R^p \lim \mathcal{F}_n = 0$ for all $p > 1$ and*
$$R^1 \lim \mathcal{F}_n = \mathrm{Coker}(\prod \mathcal{F}_n \to \prod \mathcal{F}_n)$$
where the map is $(x_n) \mapsto (x_n - f(x_{n+1}))$.
7. *If (K_n) is an inverse system of objects of $D(\mathcal{O})$, then there are short exact sequences*
$$0 \to R^1 \lim H^{p-1}(K_n) \to H^p(R \lim K_n) \to \lim H^p(K_n) \to 0$$

Proof. Proof of (1). If the sequence is exact, then evaluating at any weakly contractible element of $\mathcal{C}$ gives an exact sequence by Lemma 090K. Conversely, assume

that $\mathcal{F}_1(U) \to \mathcal{F}_2(U) \to \mathcal{F}_3(U)$ is exact for all $U \in \mathcal{B}$. Let V be an object of $\mathcal{C}$ and let $s \in \mathcal{F}_2(V)$ be an element of the kernel of $\mathcal{F}_2 \to \mathcal{F}_3$. By assumption there exists a covering $\{U_i \to V\}$ with $U_i \in \mathcal{B}$. Then $s|_{U_i}$ lifts to a section $s_i \in \mathcal{F}_1(U_i)$. Thus s is a section of the image sheaf $\mathrm{Im}(\mathcal{F}_1 \to \mathcal{F}_2)$. In other words, the sequence $\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3$ is exact.

Proof of (2). This holds by Lemma 0D64 with $d = 0$.

Proof of (3). Let $(\mathcal{F}_n)$ be a system as in (2) and set $\mathcal{F} = \lim \mathcal{F}_n$. If $U \in \mathcal{B}$, then $\mathcal{F}(U) = \lim \mathcal{F}_n(U)$ surjects onto $\mathcal{F}_1(U)$ as all the transition maps $\mathcal{F}_{n+1}(U) \to \mathcal{F}_n(U)$ are surjective. Thus $\mathcal{F} \to \mathcal{F}_1$ is surjective by Sites, Definition 00WM and the assumption that every object has a covering by elements of $\mathcal{B}$.

Proof of (4). Let $\mathcal{F}_{i,1} \to \mathcal{F}_{i,2} \to \mathcal{F}_{i,3}$ be a family of exact sequences of $\mathcal{O}$-modules. We want to show that $\prod \mathcal{F}_{i,1} \to \prod \mathcal{F}_{i,2} \to \prod \mathcal{F}_{i,3}$ is exact. We use the criterion of (1). Let $U \in \mathcal{B}$. Then

$$(\prod \mathcal{F}_{i,1})(U) \to (\prod \mathcal{F}_{i,2})(U) \to (\prod \mathcal{F}_{i,3})(U)$$

is the same as

$$\prod \mathcal{F}_{i,1}(U) \to \prod \mathcal{F}_{i,2}(U) \to \prod \mathcal{F}_{i,3}(U)$$

Each of the sequences $\mathcal{F}_{i,1}(U) \to \mathcal{F}_{i,2}(U) \to \mathcal{F}_{i,3}(U)$ are exact by (1). Thus the displayed sequences are exact by Homology, Lemma 060K. We conclude by (1) again.

Proof of (5). Follows from (4) and (slightly generalized) Derived Categories, Lemma 07KC.

Proof of (6) and (7). We refer to Section 0940 for a discussion of derived and homotopy limits and their relationship. By Derived Categories, Definition 08TC we have a distinguished triangle

$$R\lim K_n \to \prod K_n \to \prod K_n \to R\lim K_n[1]$$

Taking the long exact sequence of cohomology sheaves we obtain

$$H^{p-1}(\prod K_n) \to H^{p-1}(\prod K_n) \to H^p(R\lim K_n) \to H^p(\prod K_n) \to H^p(\prod K_n)$$

Since products are exact by (4) this becomes

$$\prod H^{p-1}(K_n) \to \prod H^{p-1}(K_n) \to H^p(R\lim K_n) \to \prod H^p(K_n) \to \prod H^p(K_n)$$

Now we first apply this to the case $K_n = \mathcal{F}_n[0]$ where $(\mathcal{F}_n)$ is as in (6). We conclude that (6) holds. Next we apply it to (K_n) as in (7) and we conclude (7) holds. $\square$